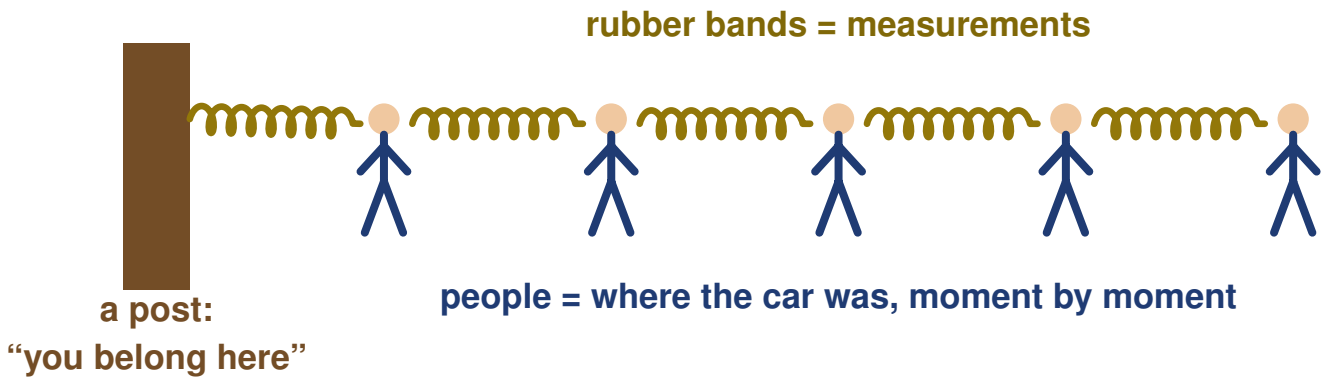
Factor Graphs and LIO-SAM

BIG IDEA

Every measurement is a rubber band, and the map settles where the pulling is shared fairly.



THE IDEA

Picture a row of people joined by rubber bands. Each band says “you two should be this far apart.” One person is also tied to a post that says “you belong here.” If one band pulls a little wrong, everyone wiggles until the pulling is shared out as fairly as possible. That picture is a **factor graph**: the people are the car’s positions over time, and the bands are measurements.

REMEMBER

- ✓ Each measurement pulls a little on the positions it connects.
- ✓ GPS is like the post: it ties the path to the real world.
- ✓ The finished path, with direction at every instant, is called the **trajectory**.

TRY THIS

Link four or five rubber bands into a chain and pin one end down. Pull the other end and watch how the stretch spreads along the whole chain.

The full module is in the Reference Book.